Discovery completeness — exhaustive InterPro / EC 6.3.- sweep

Out-of-tree audit: every reviewed ATP-dependent ligase, filtered by target chemistry, vs what the agent surfaced.

12,456
reviewed EC 6.3.- ligases (Swiss-Prot universe)

↓

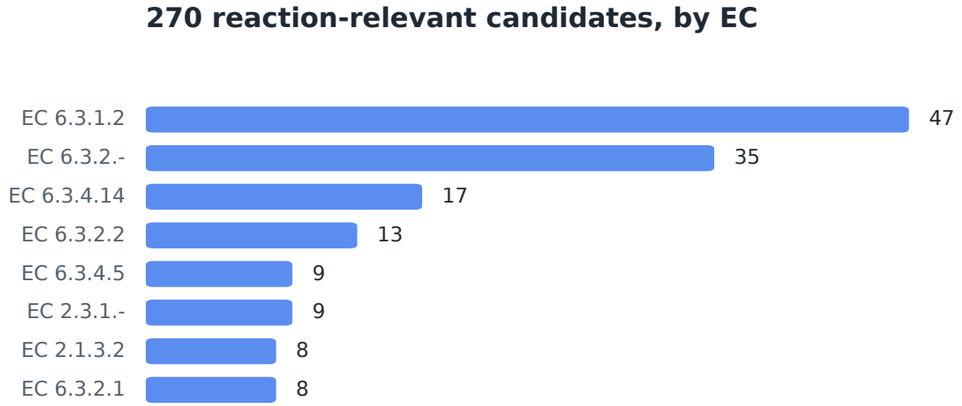
270
pass reaction-anchored chemistry filter (free diacid + amine ligation)

↓

51 / 1,858
families surveyed / candidates discovered by the agent

↓

38 → 20
ranked → handed to structural modeling



Recall ceiling (stated honestly)

The agent covers the reviewed universe — but the amide-synthetase ground truth SfaB (A0A2H4T912) is UNREVIEWED (TrEMBL, no EC), so it lies OUTSIDE this entire sweep. No reviewed-DB method recovers it; reaching it needs the literature/unannotated channel.